Schur Quotients Force ℓ_1 in Lopez-Abad Superspaces

Run fa_banach_001

2026-06-28

Status. `partial_result_likely_valid`. This packet proves a narrow obstruction for the construction route around arXiv:1210.5728. It is not a solution of the full nonseparable Bourgain–Delbaen problem.

Source target. In the introduction of Lopez-Abad's *A Bourgain-Pisier construction for general Banach spaces*, arXiv:1210.5728, the author notes that nothing is known about removing separability in the Freeman–Odell–Schlumprecht embedding theorem, or even in the more basic problem of whether a nonseparable Bourgain–Delbaen space exists, i.e. a nonseparable $\mathcal{L}_\infty$-space containing no isomorphic copies of c_0 or ℓ_1.

Idea of the proof. The quotient in the Lopez-Abad/Bourgain–Pisier extension method has the Schur property. Infinite-dimensional Schur spaces necessarily contain ℓ_1, and ℓ_1 lifts through quotient maps by its projectivity. Therefore a Schur-quotient extension cannot avoid ℓ_1 in the total space.

Theorem. *Let $q : Y \to Q$ be a quotient map between Banach spaces. If Q is infinite-dimensional and has the Schur property, then Y contains a subspace isomorphic to ℓ_1.*

Proof. First, Q contains ℓ_1. Indeed, choose a bounded sequence in Q with no norm-convergent subsequence; by passing to a subsequence, assume it is separated. If this sequence had no subsequence equivalent to the unit vector basis of ℓ_1, Rosenthal's ℓ_1 theorem would give a weakly Cauchy subsequence. The difference sequence of that weakly Cauchy subsequence is weakly null, hence norm null by the Schur property, contradicting separation. Thus some subspace $E \subseteq Q$ is isomorphic to ℓ_1.

Let $A : \ell_1 \to Q$ be an isomorphic embedding with range E. Since ℓ_1 is projective, A lifts through q: for example, for each unit vector e_n choose $y_n \in Y$ with $q(y_n) = Ae_n$ and $\|y_n\| \leq C\|Ae_n\|$, where C is a fixed quotient-open-mapping constant. Then

$$U((a_n)) = \sum_{n=1}^{\infty} a_n y_n$$

defines a bounded operator $U : \ell_1 \to Y$ and $qU = A$. Since A is bounded below and q is bounded, U is bounded below as well:

$$\|Ua\| \geq \frac{\|Aa\|}{\|q\|} \qquad (a \in \ell_1).$$

Hence $U[\ell_1]$ is a subspace of Y isomorphic to ℓ_1. □

Corollary. *Any Lopez-Abad / nonseparable Bourgain–Pisier $\mathcal{L}_\infty$-space Y whose canonical quotient Y/X is infinite-dimensional with the Schur property contains an isomorphic copy of ℓ_1. Consequently this construction class cannot provide a nonseparable $\mathcal{L}_\infty$-space containing neither c_0 nor ℓ_1.*

Proof. Apply the theorem to the quotient map $Y \to Y/X$. Lopez-Abad's construction is designed so that Y/X has the Schur property; the later analysis in Patri, arXiv:2604.11832, records the nonseparable Bourgain–Pisier Lopez-Abad class as having infinite-dimensional Schur quotient. □

Verification and limitations. The proof uses only Rosenthal's ℓ_1 theorem, the Schur property, and the elementary projectivity of ℓ_1. The conclusion is route-specific: it rules out Schur-quotient Bourgain–Pisier superspaces as candidates for the "no c_0, no ℓ_1" nonseparable Bourgain–Delbaen problem, but it does not rule out other nonseparable $\mathcal{L}_\infty$-space constructions.

Search notes. Local index searches for arXiv:1210.5728, Bourgain–Pisier, Bourgain–Delbaen, nonseparable $\mathcal{L}_\infty$, c_0, and ℓ_1 found no exact prior packet for this obstruction. A web search on 2026-06-28 found Patri, arXiv:2604.11832, concerning separable quotients of nonseparable Bourgain–Pisier spaces, but no full solution of the no-c_0, no-ℓ_1 nonseparable Bourgain–Delbaen problem.

References

[1] J. Lopez-Abad, *A Bourgain-Pisier construction for general Banach spaces*, arXiv:1210.5728.

[2] K. Patri, *Mazur's Separable Quotient Problem for Nonseparable Bourgain-Pisier $\mathcal{L}_\infty$-Spaces*, arXiv:2604.11832.

[3] H. P. Rosenthal, *A characterization of Banach spaces containing ℓ_1*, Proc. Nat. Acad. Sci. U.S.A. 71 (1974), 2411–2413.